

High-Speed Interconnects in Circuit Simulation

Theory, Matrix Structure, and Physical Interpretation

Abstract

This document presents the theory and matrix structure of high-speed interconnects in the context of the circuit simulation. It focuses on the standard per-unit-length (PUL) matrix formulation for multiconductor transmission lines (MTLs), following Clayton R. Paul's *Analysis of Multiconductor Transmission Lines* [1]. The approach is based on representing the distributed electrical properties of the line using per-unit-length matrices. For a uniform line with n signal conductors and a reference conductor (conductor 0), the coupled Telegrapher equations describe voltage and current along the line under the transverse electromagnetic (TEM) or quasi-TEM wave propagation approximation. The model is specified by four $n \times n$ matrices: inductance $\mathbf{L}$, resistance $\mathbf{R}$, conductance $\mathbf{G}$, and capacitance $\mathbf{C}$. This document gives the governing equations, physical interpretations, and standard matrix forms.

1. Multiconductor Transmission Line (MTL) System

Consider a uniform multiconductor transmission line consisting of n signal conductors and an additional conductor, designated conductor 0 (i.e., the reference conductor or ground conductor), that serves as the common reference (return) conductor [1]. The line geometry and surrounding medium are assumed uniform along the longitudinal propagation axis z . Figure ?? illustrates the physical structure of a sample multiconductor interconnect with signal traces and a ground plane.

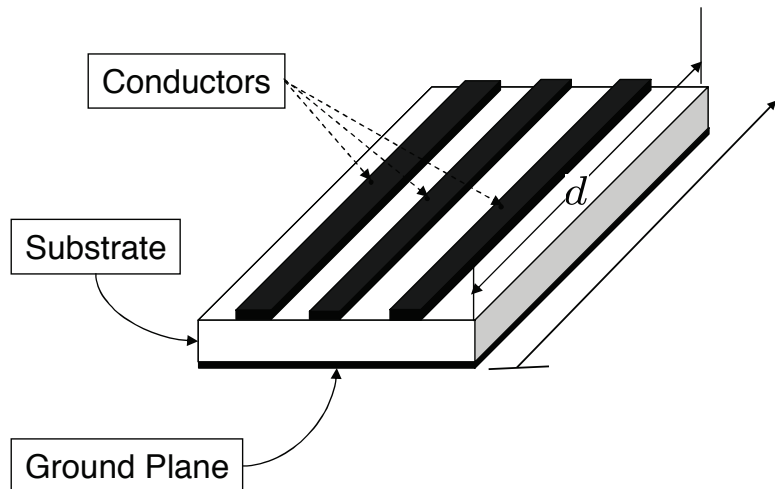


Figure 1: A physical representation for a high-speed uniform multiconductor interconnect structure with 3 signal conductors and a reference conductor (conductor 0).

The high-speed interconnect networks have their physics grounded in the theory of electromagnetic propagation, which is often described by Maxwell's equations. Under the transverse electromagnetic

(TEM) or quasi-TEM wave propagation mode [1], the voltages and currents on those network are properly described as functions of time t and a spatial variable z that denotes the spatial distance along the line.

Figure 2 illustrates a circuit schematic representing a network of n signal conductors with $2n$ ports.

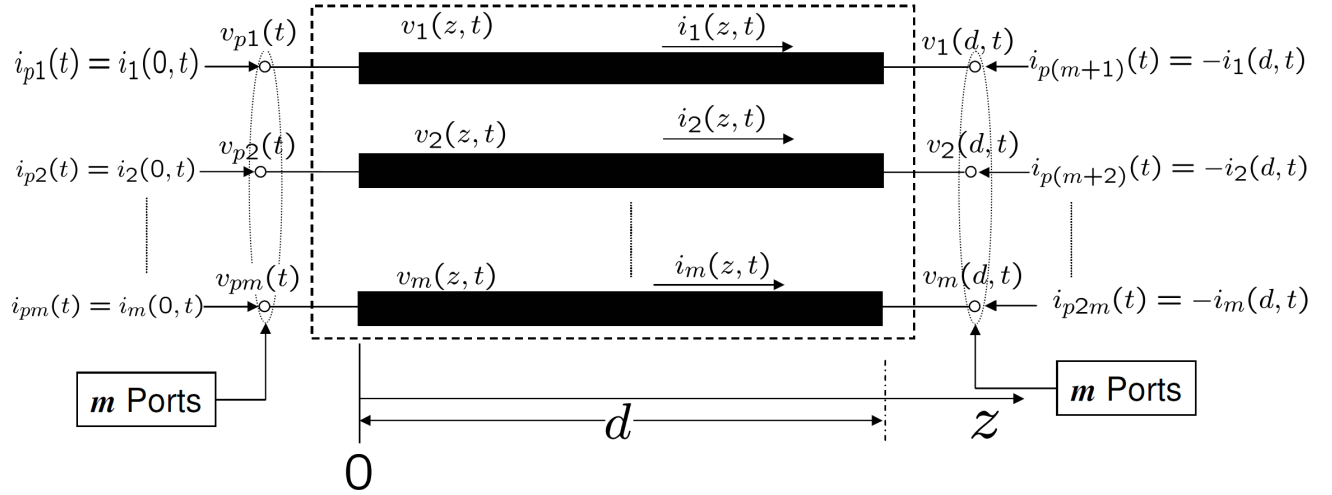


Figure 2: A schematic representation for the high-speed interconnect structure shown in Figure 1.

The line voltages of the n signal conductors with respect to the reference conductor are represented by the $n \times 1$ voltage vector $\mathbf{V}(z, t)$, and the signal conductor currents are represented by the $n \times 1$ current vector $\mathbf{I}(z, t)$. These are defined in (1) as

$$\mathbf{V}(z, t) = \begin{bmatrix} v_1(z, t) \\ v_2(z, t) \\ \vdots \\ v_n(z, t) \end{bmatrix}, \quad \mathbf{I}(z, t) = \begin{bmatrix} i_1(z, t) \\ i_2(z, t) \\ \vdots \\ i_n(z, t) \end{bmatrix}. \quad (1)$$

The total return current flowing through the reference conductor is given by Kirchhoff's Current Law:

$$i_0(z, t) = - \sum_{i=1}^n i_i(z, t).$$

As shown above, in this high-speed network, voltages and currents are no longer localized to a certain node or a certain branch as in the case of lumped elements, but are rather distributed along the lines as a function of both time t and distance z . Thus, the constitutive relation between the voltage and currents, in this case, is no longer an algebraic or simple differential relation (initial value problem–IVP), but rather takes the form of a set of Partial Differential Equations (PDE) as a boundary value problem (BVP), in which derivatives of the voltages and currents with respect to t and z appear.

2. Coupled Multiconductor Telegrapher Equations

The spatial distribution and temporal dynamics of the voltage and current vectors along the multiconductor line are governed by the coupled Telegrapher equations in matrix differential form [1]:

$$\frac{\partial \mathbf{V}(z, t)}{\partial z} = -\mathbf{R}\mathbf{I}(z, t) - \mathbf{L} \frac{\partial \mathbf{I}(z, t)}{\partial t}, \quad (2a)$$

$$\frac{\partial \mathbf{I}(z, t)}{\partial z} = -\mathbf{G}\mathbf{V}(z, t) - \mathbf{C} \frac{\partial \mathbf{V}(z, t)}{\partial t}. \quad (2b)$$

Using the $e^{j\omega t}$ phasor convention, with angular frequency ω and phasors $\hat{\mathbf{V}}(z)$ and $\hat{\mathbf{I}}(z)$, the equations become:

$$\frac{d\hat{\mathbf{V}}(z)}{dz} = -\mathbf{Z}\hat{\mathbf{I}}(z) = -(\mathbf{R} + j\omega\mathbf{L})\hat{\mathbf{I}}(z), \quad (3a)$$

$$\frac{d\hat{\mathbf{I}}(z)}{dz} = -\mathbf{Y}\hat{\mathbf{V}}(z) = -(\mathbf{G} + j\omega\mathbf{C})\hat{\mathbf{V}}(z). \quad (3b)$$

Here, $\mathbf{Z} = \mathbf{R} + j\omega\mathbf{L}$ is the $n \times n$ per-unit-length series-impedance matrix, and $\mathbf{Y} = \mathbf{G} + j\omega\mathbf{C}$ is the $n \times n$ per-unit-length shunt-admittance matrix. The PUL units are Ω/m for $\mathbf{R}$, H/m for $\mathbf{L}$, S/m for $\mathbf{G}$, and F/m for $\mathbf{C}$.

All four matrices are per-unit-length quantities. The time-domain equations assume frequency-independent parameters; practical conductor and dielectric models may make the effective parameters frequency dependent.

3. Longitudinal Parameter Matrices (L and R)

The longitudinal parameters model the magnetic flux linkages and ohmic losses associated with current loops formed by the signal conductors and the reference conductor:

- **Per-Unit-Length Inductance Matrix, $\mathbf{L}$** ($n \times n$, symmetric positive definite for independent circuits): The inductance matrix contains the individual per-unit-length self-inductances, l_{ii} , of the circuits formed between conductor i and reference conductor 0 (i.e., ground), and the per-unit-length mutual inductances, $l_{ij} = l_{ji}$, between circuits i and j , as

$$\mathbf{L} = \begin{bmatrix} l_{11} & l_{12} & \cdots & l_{1n} \\ l_{21} & l_{22} & \cdots & l_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ l_{n1} & l_{n2} & \cdots & l_{nn} \end{bmatrix}. \quad (4)$$

- **Per-Unit-Length Resistance Matrix, $\mathbf{R}$** ($n \times n$, symmetric positive definite when all signal-conductor resistances are positive): The resistance matrix accounts for the finite conductivity of the conductors. The diagonal entries ($r_i + r_0$) represent the total round-trip loop resistance for circuit i consisting of signal conductor resistance r_i and the common return conductor resistance r_0 . The off-diagonal entries r_0 represent the common-impedance resistive coupling due to shared return current in conductor 0:

$$\mathbf{R} = \begin{bmatrix} r_1 + r_0 & r_0 & \cdots & r_0 \\ r_0 & r_2 + r_0 & \cdots & r_0 \\ \vdots & \vdots & \ddots & \vdots \\ r_0 & r_0 & \cdots & r_n + r_0 \end{bmatrix}. \quad (5)$$

where $\mathbf{V}$ and $\mathbf{I}$ are defined by (1).

4. Transverse Parameter Matrices (G and C)

The transverse parameters represent the conduction and displacement currents flowing across the dielectric medium in the transverse cross-sectional plane:

- **Per-Unit-Length Conductance Matrix, $\mathbf{G}$** ($n \times n$, symmetric): Represents transverse conduction current (dielectric leakage losses) flowing between the conductors in the transverse plane as

$$\mathbf{G} = \begin{bmatrix} \sum_{k=1}^n g_{1k} & -g_{12} & \cdots & -g_{1n} \\ -g_{21} & \sum_{k=1}^n g_{2k} & \cdots & -g_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -g_{n1} & -g_{n2} & \cdots & \sum_{k=1}^n g_{nk} \end{bmatrix}. \quad (1)$$

Let $g_{ij} = g_{ji} \geq 0$ be the conductance between signal conductors i and j , and let $g_{i0} \geq 0$ be the conductance between signal conductor i and the reference. Then

$$\mathbf{G} = [G_{ij}], \quad G_{ij} = \begin{cases} g_{i0} + \sum_{\substack{k=1 \\ k \neq i}}^n g_{ik}, & i = j, \\ -g_{ij}, & i \neq j. \end{cases} \quad (6)$$

- **Per-Unit-Length Capacitance Matrix, $\mathbf{C}$** ($n \times n$, symmetric positive definite for a well-posed cross-section referenced to conductor 0): Represents transverse displacement current (electric-field coupling) flowing between the conductors in the transverse plane as

$$\mathbf{C} = \begin{bmatrix} \sum_{k=1}^n c_{1k} & -c_{12} & \cdots & -c_{1n} \\ -c_{21} & \sum_{k=1}^n c_{2k} & \cdots & -c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -c_{n1} & -c_{n2} & \cdots & \sum_{k=1}^n c_{nk} \end{bmatrix}. \quad (2)$$

Let $c_{ij} = c_{ji} \geq 0$ be the partial capacitance between signal conductors i and j , and let $c_{i0} \geq 0$ be the partial capacitance between signal conductor i and the reference. Then

$$\mathbf{C} = [C_{ij}], \quad C_{ij} = \begin{cases} c_{i0} + \sum_{\substack{k=1 \\ k \neq i}}^n c_{ik}, & i = j, \\ -c_{ij}, & i \neq j. \end{cases} \quad (7)$$

Matrix structure. For both transverse matrices, each diagonal entry is the sum of all branches incident on that conductor, including its branch to conductor 0; each off-diagonal entry is the negative of the mutual branch parameter. Thus the matrices have nonpositive off-diagonal entries and are diagonally dominant. The conductance matrix is positive semidefinite, and is positive definite when every connected component of the transverse conductance network has a path to the reference. The capacitance matrix is positive definite for a well-posed cross-section referenced to conductor 0.

4.1 Macromodelling of High-Speed Interconnects

One class of typical approaches used to circumvent the problem of including high-speed interconnects in the circuit formulation is based on deriving a time-domain stamp for the interconnect element. Such models can be developed, but usually at the cost of compactness. These approaches, collectively known under the term “macromodelling”, seek to convert the PDE into a time-domain ODE that can be easily stamped into the modified nodal analysis (MNA) matrix formulation [2, 3].

Examples of these techniques are the Matrix Rational Approximation (MRA) [4, 5] of the exponential function and delay-based approaches such as Method of Characteristics (MoC) [6] and the DEPACT algorithm [7]. Macromodelling, used in that sense, solves the issue of stamping the high-speed interconnect into the circuit formulation, but at the cost of compactness as it produces a large network of extra circuit elements. The goal of using MOR in high-speed interconnects is to address the complexity of simulating the circuit with large macromodels. The next subsection presents the macromodelling approach based on lumped segmentation by discretizing the PDE along the spatial dimension z .

4.2 Time-Domain macromodelling Based on Discretization

Discretization techniques represent a very straightforward approach to overcome the above difficulties and incorporate high-speed interconnects in circuit simulators. Following the central idea of these techniques, one first needs to introduce the parameters of the line, which are the resistance (R), inductance (L), conductance (G), and capacitance (C) per unit length. In the case of an interconnect with m conductors, those parameters are $m \times m$ matrices.

Discretization proceeds by dividing the line into segments of length Δz , chosen to be a small fraction of the shortest wavelength in the driving signal (the shortest wavelength is obtained by dividing the speed of light in the medium of the interconnect by the highest frequency in the propagating signal). If the length of each segment is electrically small (i.e., compared to the shortest wavelength), then each segment can be replaced by the model shown in Figure 3.

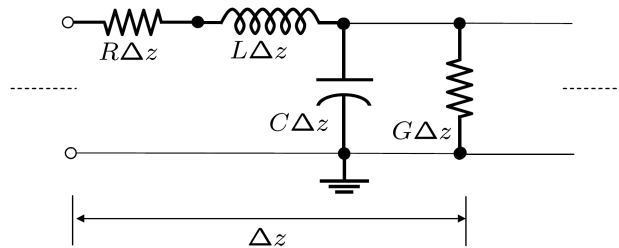


Figure 3: Modeling a segment of a single-conductor transmission line using lumped circuit elements.

It is of practical interest to know how many of these segments are required to reasonably approximate the interconnect. For illustration consider a lossless line, i.e., $R = 0$ and $G = 0$, with only LC elements

which can be viewed as a low pass filter. For a reasonable approximation, this filter must pass at least some multiples of the highest frequency $f_{\max}$ of the propagating signal (say ten times, $f_0 \geq 10f_{\max}$). To relate these parameters, we make use of the 3-dB passband frequency of the LC filter given by [8]

$$f_0 = \frac{1}{\pi\sqrt{LdCd}} = \frac{1}{\pi\tau d} \quad (3)$$

where d is the length of the line and $\tau = \sqrt{LC}$ represents the delay per-unit-length (p.u.l). Typically, we set $f_{\max} = 0.35/t_r$, where t_r is the rise time of the signal. Using (3), we can express the relation $f_0 \geq 10f_{\max}$ in terms of the delay of the line and the rise time as $1/(\pi\tau d) \geq 10 \times 0.35/t_r$ or $t_r \geq 3.5(\pi\tau d) \approx 10\tau d$. In other words, the delay allowed per segment is approximately $t_r/10$. Hence the total number of segments (P) needed to represent the total delay of τd is given by

$$P = \tau d/(t_r/10) = 10\tau d/t_r. \quad (4)$$

As an example, consider a digital signal with rise time $t_r = 0.2$ ns propagating on a lossless wire of length 10 cm with a p.u.l. delay of 70.7 ps, which can be represented by a distributed model with Per-Unit-Length (p.u.l) parameters of $L = 5$ nH/cm and $C = 1$ pF/cm. If the same transmission line were to be represented by lumped segments, one needs $P \approx 35$ segments.

One of the major drawbacks of the above approach is that it requires a large number of sections, especially for circuits with many multiple conductors, high operating speeds, and sharper rise times. This leads to large circuit sizes and the simulation becomes CPU inefficient.

The approaches that are often used to circumvent this difficulty, i.e., to derive a time-domain stamp for the interconnect element, can be developed, but usually at the cost of compactness. Those approaches, which are collectively known under the term “macromodelling”, seek to convert the PDE into a time-domain ODE that can be easily stamped into the modified nodal analysis formulation.

Macromodelling techniques typically approach the problem by discretizing the PDE along the spatial dimension z , approximating the z derivative using an appropriate approximation operator. Examples of these techniques are the Matrix Rational Approximation (MRA) [4, 5] of the exponential function and delay-based approaches such as Method of Characteristics (MoC) [6] and the DEFACT algorithm [7]. Macromodelling, used in that sense, solves the issue of stamping the high-speed interconnect in the circuit formulation, but at the cost of compactness as it produces a large network of extra circuit elements. The goal of using MOR in high-speed interconnects is to address the complexity of simulating the circuit with large macromodels.

The next subsection presents the macromodelling based on lumped segmentation, while the rest of the chapter will review the application of various MOR approaches used to reduce the complexity of the models.

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